AI USAGE & PROMPT LOG

(GitHub Copilot – VS Code)

Note: The following log records *human-visible prompts and responses* from GitHub Copilot Chat and inline usage. Internal system prompts and model context are not accessible due to tool limitations. All entries were manually recorded and verified by the author.

Prompt ID: 01

Tool: GitHub Copilot Chat (VS Code)
Model: Proprietary (GitHub Copilot)

Prompt

"Build me a DMA which is arbitrated-ended, error-detecting, and low power. Do not change any existing files. Build new ones that integrate into this. Make the most efficient and fast DMA."

AI Response Summary

- Proposed a new DMA peripheral module tqvp_dma.v
- Exposed:
  - Memory-mapped register interface
  - Optional master bus interface (m_* signals)
  - Low-power idle indication
- Provided integration notes and register map (dma_README.md)
- Requested clarification on:
  - Target bus interface for DMA master ports
  - Whether to auto-instantiate DMA into peripherals.v

Human Action

- Chose to proceed with explicit peripheral and bus selection in follow-up prompts
- Did not allow modification of existing files without review

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Prompt ID: 02

Tool: GitHub Copilot Chat (VS Code)

Prompt

"Choose the peripherals and bus to implement peripheral-to-memory, memory-to-memory, and peripheral-to-peripheral transfers. Add a testbench to illustrate register programming and a README for testing."

AI Response Summary

- Identified existing memory/master-style interfaces in the repository
- Added:
  - tqvp_dma.v (DMA implementation)
  - tb_dma.v (Verilog testbench)
  - README files documenting:
    - Integration
    - Register programming
    - Test execution with Icarus Verilog
- Testbench covered:
  - Memory → Memory
  - Peripheral → Memory
  - Peripheral → Peripheral

Human Action

- Reviewed architecture and test strategy
- Continued iterative validation through simulation feedback

Prompt ID: 03

Tool: GitHub Copilot Chat (VS Code)

Prompt

"What all can the entire project do now? How can I test this?"

AI Response Summary

- Summarized current system capabilities:
  - DMA peripheral with:
    - Memory-mapped control
    - Channel selection
    - Interrupt on completion
    - Error detection
    - Low-power idle signal
  - Testbench scenarios and expected outputs
- Provided:
  - Instructions for running tb_dma.v

    - Integration options for SoC-level wiring
- Listed next possible enhancements (scatter/gather, burst transfers, etc.)

Human Action

- Followed test instructions
- Proceeded to verification and correctness review

Prompt ID: 04

Tool: GitHub Copilot Chat (VS Code)

Prompt

"Check the entire code and ensure everything is correct for the DMA. Give instructions to test the code."

AI Response Summary

- Performed iterative debugging of the testbench:
    - Fixed Verilog syntax constraints
    - Replaced unsupported constructs in Icarus Verilog
    - Identified and corrected DMA address sequencing bug
- Updated DMA logic:
    - Introduced proper read/write pipelining
    - Ensured data/address stability across cycles
- Final result:
    - All testbench scenarios passed successfully

Verified Deliverables

- tqvp_dma.v — production-ready DMA module
- tb_dma.v — full testbench (all scenarios passing)
- Verified features:
    - Arbitrated channel selection
    - Error detection
    - End-of-transfer interrupt
    - Low-power idle signal
    - 32-bit word-aligned transfers

Human Action

- Reviewed fixes
- Accepted final DMA implementation after successful verification

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Prompt ID: 05

Tool: GitHub Copilot Chat (VS Code)

Prompt

"We have to pass the benchmarks according to this: (attached test.py)"

AI Response Summary

- Created cocotb-based DMA verification tests aligned with the provided TinyQV test harness
- Added:
    - test_dma.py (cocotb tests)
    - README documenting cocotb setup and execution
- Re-verified:
    - Verilog testbench compatibility
    - Cocotb test coverage
- Generated final documentation and verification checklist

Final AI-Assisted Output Summary

Functional Capabilities

- Arbitrated DMA (4 selectable channels)
- Supported transfer modes:
    - Memory → Memory
    - Peripheral → Memory
    - Peripheral → Peripheral
- Error detection with halt-on-failure
- End-of-transfer interrupt support
- Low-power idle indication
- Pipelined operation (~1 word per 2 cycles)

Files Produced

Implementation

- tqvp_dma.v

Verification

- tb_dma.v
- tb_dma_simple.v
- test_dma.py

Documentation

- dma_README.md
- TEST_DMA_README.md
- DMA_IMPLEMENTATION_GUIDE.md
- DMA_VERIFICATION_CHECKLIST.md

Human Contribution Statement

- All AI-generated code was:
  - Reviewed
  - Debugged
  - Modified where required
- Architectural decisions, verification acceptance, and final integration responsibility remain with the author.
- Timing closure considerations, register placement decisions, and compatibility with TinyQV bus assumptions were evaluated manually by the author.

Clarification on Copilot Usage
GitHub Copilot was used in both **inline code completion** and **chat-assisted design discussion** modes. Inline suggestions (e.g., signal naming, boilerplate always blocks, register definitions) are not associated with explicit prompts and therefore cannot be logged verbatim. Only human-visible Copilot Chat interactions and high-level design prompts are recorded here.